

Event-Camera 3D Reconstruction Pipeline

Objective. Reconstruct a 3D scene from event-camera data captured with a LUCID event camera fusing in synchronized RGB-D.

Phase 1 - Validation baseline. Convert the event stream to intensity frames (E2VID), recover camera poses and a sparse point cloud with COLMAP structure-from-motion, and run standard 3D Gaussian Splatting. This confirms the data is geometrically consistent and establishes a quantitative floor to beat.

Phase 2 - Custom pipeline. Combine the complementary strengths of two methods. EventSplat supplies the front end, event-to-video initialization, SfM, and spline-interpolated continuous poses, solving the cold-start and pose problems. Event-3DGS supplies the back end, photovoltage-based denoising, an event-supervision loss, and joint optimization of sensor parameters, giving robustness on real, noisy captures.

Phase 3 - RGB-D fusion. A rigidly co-mounted RGB-D camera on the same trajectory adds metric scale, dense geometric initialization, and color, while events cover the fast-motion and low-light gaps where depth sensors fail. Temporal sync relies on a hardware trigger that logs timestamped trigger-events directly in the event stream (or PTP for a shared clock), with cross-sensor extrinsics calibrated via e2calib.

Currently planning out phase 2.

